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Faculdade de Tecnologia
Departamento de Engenharia Mecânica

Gated Data-Space Inversion for Reservoir Prediction

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DISSERTAÇÃO DE MESTRADO
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Previsão de Reservatórios usando Inversão do Espaço de Dados Ponderada

Thiago Tomás de Paula

Dissertação de Mestrado submetida ao Programa de Pós-Graduação em Ciências Mecânicas como requisito parcial para obtenção do grau de Mestre.

Orientador: Prof. Dr. Eugênio Libório Feitosa Fortaleza

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To all fascinated with almost understanding things

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*“Engineers use science to solve their problems if the science is available.
But available or not, the problem must be solved,
and whatever form the solution takes under these conditions
is called engineering.”
(from Shigley’s Mechanical Engineering Design)*

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One of this dissertation's case studies is a real oilfield in the Brazilian Post-Salt, whose data are sensitive. Petrobras SA has given permission to use, and to some extent, disclose, data related to this field for the purposes of this work.

Abstract

Accurate prediction of reservoir performance under uncertainty is essential for decision-making in geological reservoir management. Data-Space Inversion (DSI) (Sun; Durlowsky, 2017) provides an attractive alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations. Recently, Lima, Emerick, and Ortiz (2020) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. Despite these advantages, DSI-ESMDA, like other ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty. Methods such as covariance localization help to ameliorate this issue, but variance underestimation is still observed in practical applications. One contributing factor to this progressive loss of variance is that small production mismatches, often dominated by measurement noise or sampling effects, may trigger repeated micro-updates that reduce ensemble diversity without improving predictive skill. This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems (Fortaleza *et al.*, 2022) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing (Donoho; Johnstone, 1994; Donoho, 1995). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The approach is evaluated on multiple synthetic reservoirs and one real field application using production rates from producers and injectors. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

Keywords: ; ; .

Resumo

Palavras-chave: ; ; .

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1 Introduction

This document is the author's master's dissertation submitted to the Mechanical Sciences Graduate Program at University of Brasília as a partial requisite in acquiring the master's degree. Remaining requisites the author has satisfied are presented in [appendix A](#).

1.1 Work overview

The economic success of oil and gas fields is increasingly contingent on the quality of reservoir management decisions, which are made under uncertainty and based on limited data. Kalman-based ensemble methods are widely used to quantify uncertainty in reservoir forecasts by adjusting reservoir models to match production data, but require vast computational resources to perform repeated reservoir simulations. In an effort to reduce the computational burden, Data-Space Inversion (DSI) ([Sun; Durlofsky, 2017](#)) was introduced as an alternative to traditional history matching by updating ensemble forecasts directly in data space, thereby avoiding repeated subsurface simulations.

[Lima, Emerick, and Ortiz \(2020\)](#) introduced DSI-ESMDA, an extension of DSI that employs the Ensemble Smoother with Multiple Data Assimilation (ES-MDA) to replace the Randomized Maximum Likelihood (RML) minimizations of the original formulation with an iterative Kalman-type update, achieving significant computational efficiency while maintaining competitive posterior forecasts. However, DSI-ESMDA, like related ensemble smoothers, remains susceptible to excessive variance loss, which can lead to underestimation of posterior uncertainty.

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. Findings on both synthetic and real reservoirs show that Gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without degrading the data match.

Gated DSI-ESMDA acts as a stabilizing extension rather than a replacement, and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The power of this contribution is that it provides a simple yet effective mean of improving the robustness of ensemble-based inversion methods without requiring additional simulations or complex modifications to existing workflows.

1.2 Background context

Prediction of future behavior of a geological reservoir given past data is a difficult inverse problem (Oliver; Chen, 2011) since satisfactorily matching a reservoir model production to past production does not guarantee future data will also be satisfactorily matched. In such cases, readjustments of model parameters is often performed, leading to further simulations and analysis. Figure 1.1 gives the general outline of a standard history matching strategy applied to a single model.

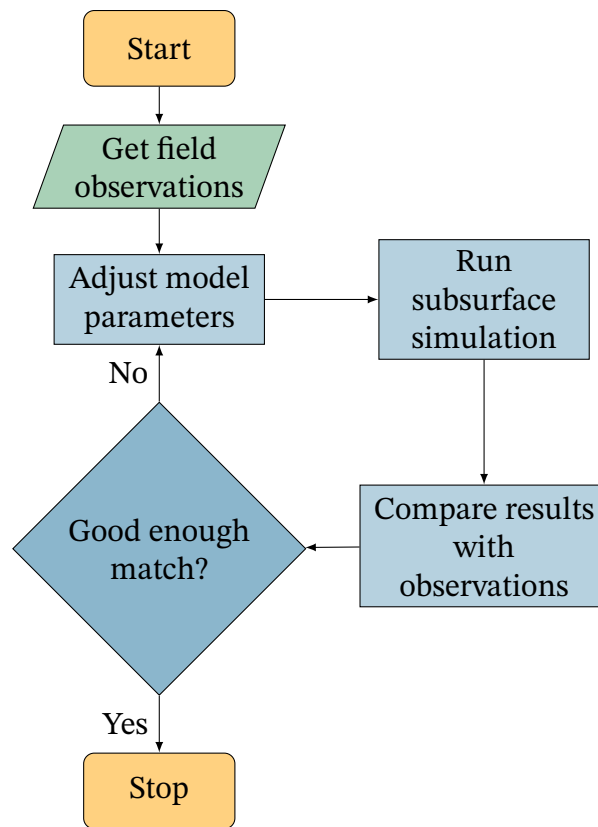


Figure 1.1 – Overview of a standard history matching strategy applied to a single model.

1.3 Research problem

Literature on DSI and DSI-ESMDA take different approaches to mitigating variance loss, but they are either computationally expensive or require careful tuning of hyperparameters. Ideally, a solution would be simple to implement, require minimal tuning, and be computationally efficient. This work proposes introducing a straightforward gating mechanism on DSI-ESMDA's innovation vector, which selectively suppresses the influence of well-matched observations while preserving the impact of significant mismatches. Gated DSI-ESMDA thus modulates the posterior ensemble variance with a single parameter, the intensity of the gating function.

1.4 Research questions

This work addresses the following research questions:

1. Can a gating mechanism be introduced into DSI-ESMDA to selectively modulate the influence of observations based on their match quality?
2. Does the proposed innovation gating reduce variance loss during assimilation and produce more robust uncertainty envelopes in the prediction phase without degrading the data match?
3. How does the performance of Gated DSI-ESMDA compare to standard DSI-ESMDA across synthetic and real reservoir case studies?

1.5 Significance

This work introduces an extension to DSI-ESMDA that modifies the effective influence of each observation through a nonlinear, misfit-dependent weighting of the innovation vector. The proposed formulation can be interpreted as introducing a smoothing influence function into the Kalman-type update, computed from normalized innovations using an error-function-based gating operator. In contrast to covariance localization, which acts on second-order statistics, innovation gating acts directly on the update term, selectively suppressing corrections associated with sufficiently-matched observations while preserving the full impact of significant mismatches. Match quality is thresholded by a gating parameter during innovation normalization. The gating methodology is inspired by adaptive filtering developed to mitigate actuator wear in closed-loop control systems ([Fortaleza et al., 2022](#)) and is conceptually related to smooth shrinkage and noise-gating strategies in signal processing ([Donoho; Johnstone, 1994](#); [Donoho, 1995](#)). Although the present work focuses on DSI, the proposed formulation is general and can be applied to standard ensemble-based model-space inversion methods, including ES-MDA and related iterative smoothers. The proposed method inherits the computational efficiency of DSI-ESMDA while providing a robust, easily tunable mechanism for modulating posterior distribution variance, setting it as a practical and effective tool for uncertainty quantification in reservoir forecasting.

1.6 Methodology overview

The approach is evaluated against the original DSI-ESMDA on two synthetic reservoirs and one real field application using measurements from producers and injectors. The synthetic reservoirs are the Egg ([Jansen et al., 2014](#)) and Olympus ([Fonseca et al., 2020](#)) ensembles, which are widely used in the literature to benchmark history matching and uncertainty quantification methods, and the real field application is a mature oil field in

the Brazilian Post-Salt. Posterior and prior ensembles are compared across several metrics, including observation coverage and mean squared error (Emerick, 2025, subsection 9.4.2), to assess the impact of innovation gating on variance loss and uncertainty quantification. Results show that innovation-gated DSI-ESMDA reduces variance loss during and produces more robust uncertainty envelopes in the prediction phase without significantly degrading the data match. Furthermore, for moderate gating parameters, posterior forecasts remain close to those obtained with standard DSI-ESMDA, indicating that the proposed method acts as a stabilizing extension rather than a replacement.

1.7 Dissertation outline

Chapter 2 presents a review of the literature on reservoir uncertainty quantification, focusing on ensemble-based methods and data-space inversion. Chapter 3 introduces the theoretical framework underpinning the proposed methodology, from the relevant elements of reservoir engineering (reservoir formation, reservoir production, and subsurface simulation), to the formulation of ensemble Kalman filtering, to the presentation compact correlation functions, and finally the mathematical formulation of DSI-ESMDA and the innovation gating mechanism. Chapter 4 details the methods used to evaluate the proposed approach, including descriptions of the synthetic and real reservoir case studies, data collection, and analysis techniques. Chapter 5 presents the results of applying innovation-gated DSI-ESMDA to the case studies, including quantitative and qualitative assessments of performance relative to standard DSI-ESMDA. Chapter 6 discusses the implications of the findings and their relationship to the existing literature. Chapter 7 summarizes the main contributions and provides final remarks on the research and future work.

2 Literature Review

2.1 Chapter outline

Aims and structure

2.2 History Matching

Subsurface simulations are computationally expensive, many methods in the literature have been proposed to avoid the adjustment loop by directly adjusting the model production curves to match past data, without updating model parameters. Notably, this does not do away with the need for a model, since the model is still used to generate the initial ensemble of production curves, so such methods are not to be taken as substitutes of standard model-based approaches, but rather as complementary techniques.

2.3 Data-Space Inversion

A Data-Space Inversion (DSI) algorithm tackles the data matching problem by directly updating an ensemble of forecasts, doing away with the update of model parameters.

Originally, DSI used RML for sampling (Sun; Durlofsky, 2017), requiring the solving of several minimization problems. DSI with Ensemble Smoother Multiple Data Assimilation (DSI-ESMDA) (Lima; Emerick; Ortiz, 2020) speeds up the original DSI process by swapping the RML by iterative, Kalman-like updates to each member of the ensemble. Recently, deep learning techniques have been used to further enhance DSI-ESMDA, with good results (Jiang; Hui; Durlofsky, 2021; Xiao *et al.*, 2023; Jiang; Durlofsky, 2024).

However, DSI-ESMDA may still make poor predictions despite good fit to past data. In the language of Bayesian statistics, the starting ensemble data are samplings of the prior distribution of reservoir production, and the ensemble data after DSI updates are samplings of the posterior distribution of reservoir production. In that sense, poor results beyond the history period means that the ensemble posterior narrows around a curve that is not representative of the reference reservoir.

While the inherent problem of history matching may be impossible to solve, it may be mitigated by allowing the posterior to spread more. In this paper, we propose a version of DSI-ESMDA that introduces variance in the posterior ensemble by adaptively and smoothly diminishing the effect of an iteration update the smaller the matching error. This preserves the benefits of DSI-ESMDA, such as physics-driven localization and efficient execution,

while equipping it with an easy to implement, statistically based mechanism for posterior ensemble adjustment that is controlled by a single, easy-to interpret hyperparameter.

2.4 Chapter summary

Key takeaways

3 Theory framework

3.1 Chapter outline

Aims and structure

3.2 Elements of reservoir engineering

3.2.1 Reservoir formation

3.2.2 Reservoir simulation

3.3 Kalman Filtering

3.4 Compact Correlation Functions

3.5 Data-Space Inversion

Let $\mathbf{d}$ denote the true production data vector of a reference reservoir across time, and let $\mathbf{d}_j$ denote the production data vector produced by simulating a realization of the reference reservoir using model parameters $\mathbf{m}_j$. For an ensemble of N_e realizations, we have N_e different such vectors, each of which dependent on its own parameters vector.

Of the original reference data vector, only part of it is used in the history matching process. Denote this vector by $\mathbf{d}_h$. Following the same pattern, $\mathbf{d}_{h,j}$ denotes the history match period data output by member j of the ensemble. Assume $\mathbf{d}_h$ is corrupted by additive measurement noise $\mathbf{e}_d \sim N(\mathbf{0}, \mathbf{C}_e)$, where $\mathbf{C}_e$ is the noise covariance matrix. The resulting vector is the observed data during the history period, denoted $\mathbf{d}_{\text{obs}}$. DSI-ESMDA updates the ensemble outputs by iterating a Kalman-like data assimilation, shown below (Lima; Emerick; Ortiz, 2020):

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)(\mathbf{d}_{\text{obs}} + \sqrt{\alpha_k} \mathbf{e}_j^k - \mathbf{d}_{h,j}^k), \quad (3.1)$$

with $1 \leq k \leq N_a$, N_a the number of assimilations DSI-ESMDA must do. Thus, $\mathbf{d}_j^k$ for $k = 1$ is the ensemble prior $\mathbf{d}_j$, and $\mathbf{d}_j^{N_a+1}$, the posterior. Vector $\mathbf{e}_j^k \sim N(\mathbf{0}, \mathbf{C}_e)$ is a random disturbance, and α_k coefficients are positive constants that satisfy $\sum_k 1/\alpha_k = 1$.

Matrices $\mathbf{R}$ and $\mathbf{K}$ are the localization and the modified Kalman gain, respectively. $\mathbf{K}$ is defined as

$$\mathbf{K}^k = \Delta \mathbf{D}^k (\Delta \mathbf{D}_h^k)^\top \left(\Delta \mathbf{D}_h^k (\Delta \mathbf{D}_h^k)^\top + \alpha_k \mathbf{C}_e \right)^{-1}, \quad (3.2)$$

with

$$\Delta \mathbf{D} = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_1 - \bar{\mathbf{d}} & \cdots & \mathbf{d}_{N_e} - \bar{\mathbf{d}} \end{bmatrix}, \quad (3.3)$$

$$\bar{\mathbf{d}} = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_j, \quad (3.4)$$

a matrix that quantifies the spread of the ensemble. $\Delta \mathbf{D}_h$ is defined analogously by constraining the above equations to the history matching period:

$$\Delta \mathbf{D}_h = \frac{1}{\sqrt{N_e - 1}} \begin{bmatrix} \mathbf{d}_{h,1} - \bar{\mathbf{d}}_h & \cdots & \mathbf{d}_{h,N_e} - \bar{\mathbf{d}}_h \end{bmatrix}, \quad (3.5)$$

$$\bar{\mathbf{d}}_h = \frac{1}{N_e} \sum_{j=1}^{N_e} \mathbf{d}_{h,j}. \quad (3.6)$$

$\mathbf{R}$ adjusts the Kalman gain, preserving correlated measures while diminishing the impact of spuriously correlated measures. As such, its entries lie in $[0, 1]$, and pointwisely multiply $\mathbf{K}$. Intuitively, a correlation is strong if measures of a same quantity were taken closer in space and/or in time, and weaker otherwise. Denoting this “distance” by δ_{uv} , where u is an index from $\mathbf{d}_j$ and v an index from $\mathbf{d}_h$, $\mathbf{R}$ is given by

$$R_{uv} = r(\delta_{uv}), \quad (3.7)$$

$$r(\delta) = \begin{cases} -\frac{1}{4}\delta^5 + \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 - \frac{5}{3}\delta^2 + 1, & 0 \leq \delta \leq 1 \\ \frac{1}{12}\delta^5 - \frac{1}{2}\delta^4 + \frac{5}{8}\delta^3 + \frac{5}{3}\delta^2 - 5\delta + 4 - \frac{2}{3}\delta^{-1}, & 1 < \delta \leq 2 \\ 0, & \delta > 2 \end{cases}, \quad (3.8)$$

$$\delta_{uv} = \sqrt{\left(\frac{\Delta x'_{uv}}{L_{x'}} \right)^2 + \left(\frac{\Delta y'_{uv}}{L_{y'}} \right)^2 + \left(\frac{\Delta t_{uv}}{T_c} \right)^2}, \quad (3.9)$$

$$\begin{bmatrix} \Delta x'_{uv} \\ \Delta y'_{uv} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \Delta x_{uv} \\ \Delta y_{uv} \end{bmatrix} \quad (3.10)$$

Δx_{uv} and Δy_{uv} are the x and y differences, respectively, between the points in space from where measure u and v were taken. In this context, measurements always come from a well, and so these differences are always distances between wells. Δt_{uv} is the difference between the time when measurement u was taken, and the time when v was taken. Note that while v is restrained to the history matching period by definition, u is not. The inclusion of Δt_{uv} guarantees that measurements far apart in time will hold little correlation, even if taken from the same well.

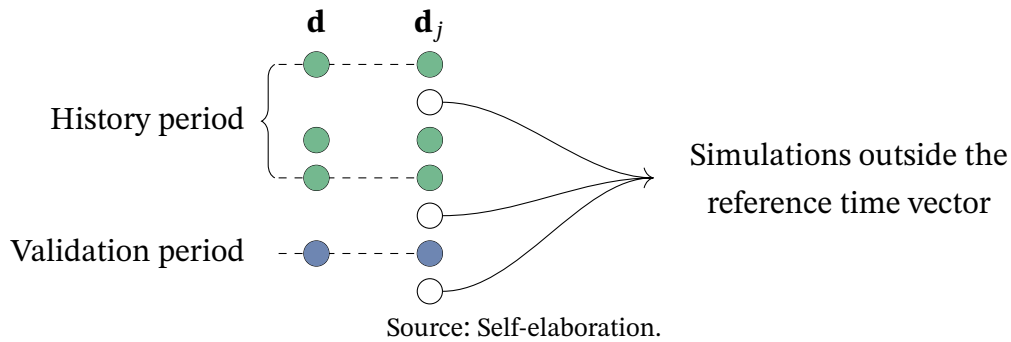
Implementation note 3.1: Index v

Data vectors stack time series of a fields' measurements, so index v in this chapter's equations specify a datum in place and time simultaneously. For example, if DSI is being applied on the oil production of two wells of a field, each with a simulation time vector of 100 elements, then for any $1 \leq j \leq N_e$ $\mathbf{d}_j$ would be 200×1 , with elements $v = 1$ through $v = 100$ the oil production history of the first well, and elements $v = 101$ through $v = 200$ the oil production history of the second well.

Implementation note 3.2: Time vectors

Note that all realization data vectors are implicitly assumed to have a synchronized time vector. Ultimately, this means simulation scheduling was the same for all realizations. Reference data, however, is under no obligation to follow this scheduling. As such, in order for [equation \(3.1\)](#) to work, the history period of the realization data is constrained to match the observed data's. This leads to an important practical requirement of the algorithm: for simulations to cover the time vector of observed data. [Figure 3.1](#) illustrates this process for data vectors with a single feature (e.g., field oil production total).

Figure 3.1 – Synchronization between realization data vector and reference data.



A rotation by θ is applied to take into account the anisotropy of reservoir permeability, and are $L_{x'}$, $L_{y'}$, T_c positive constants that regulate the critical points of differences in x , y and t separately. By critical, it is understood the inflection point $\delta = 1$ in $r(\delta)$. As illustrated by [figure 3.3c](#), the greater the critical value, the more lax the localization, allowing more weakly correlated changes to hold more impact.

Figure 3.2 – Illustration of the localization applied in several different measurements. *Left panel:* Localization between measurements from the same well but from different times. *Center panel:* Localization between measurements from the different wells but at the same times. *Right panel:* Localization between measurements from the different wells at different times.

Localization greatly improves the quality of the posterior, but if the prior zeroes in on an average that is too different from the reference, it is hard to adjust it in order to inject some variance on the collapsed curves. This is partly the reason behind the random disturbance in: injecting variance on the ensemble fit.

Even when the localization $\mathbf{R}$ is well tuned (i.e., θ , $L_{x'}$, $L_{y'}$ and T_c generate adequate posteriors), DSI-ESMDA may suffer the inherent history matching problem of making poor predictions despite good fit to past data. The ensemble posterior narrows around a curve that is not representative of the reservoir.

The inherent problem of history matching may be impossible to solve, but it may be mitigated by allowing the posterior to spread more. We propose doing so by adaptively adjusting the importance of the history mismatch when running the update iterations:

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + (\mathbf{R} \circ \mathbf{K}^k)[\alpha_j^k \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^k)], \quad (3.11)$$

$$\alpha_{j,v}^k = \text{erf} \left(\frac{1}{\lambda \sqrt{2\sigma_v^2}} |(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^k)_v| \right) \quad (3.12)$$

The equation above is based on the work of [Fortaleza et al. \(2022\)](#), and a detailed derivation is provided in [appendix B](#). The smaller the mismatch of a measurement relative to the noise at that measure, the smaller the update. Constant $\lambda \geq 1$ inflates the deviation, further softening the Kalman-like update. At the extreme ($\lambda \rightarrow \infty$ or $\mathbf{d}_{\text{obs}} = \mathbf{d}_{h,j}^k$), $\alpha_j^k = \mathbf{0}$ and the DSI update is fully undone.

Figure 3.3 – Important functions for understanding the proposed Data-Space Inversion algorithm.

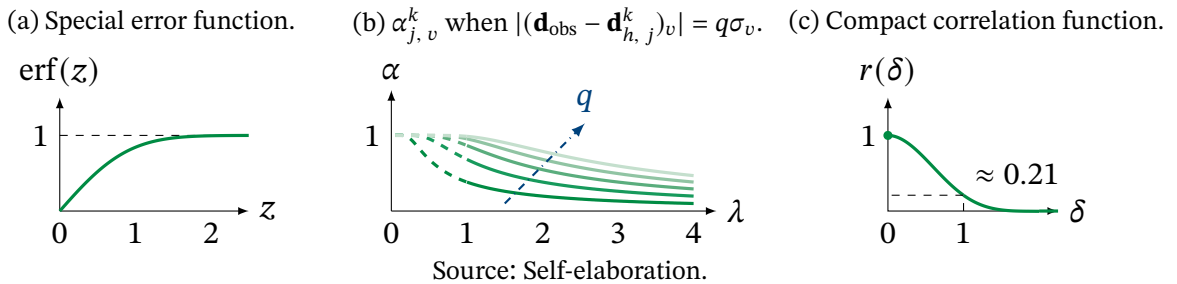


Figure 3.4 – Illustration of the effect of λ on the posterior ensemble for fixed $\mathbf{R}$ and N_a . Yellow lines are the values of $\alpha_j^{N_a+1}$ across time. *Left panel*: DSI-ESMDA prior and posterior ensembles. *Center panel*: DSI-ESMDA with $\lambda = 1$ filtering. *Right panel*: DSI-ESMDA with $\lambda = 5$ filtering. Posterior gains variance as λ grows.

Source: Self-elaboration.

3.6 Chapter summary

Key takeaways

4 Methods

4.1 Chapter outline

Aims and structure

4.2 Reservoir case studies

Three reservoirs were used as case studies for this work, two of which are synthetic reservoirs and one is a real reservoir. This section presents them, highlighting their most significant structural and geological characteristics within the scope of this work. However, as information regarding the real reservoir is sensitive, it is only briefly discussed.

The choice of these reservoirs was based on the availability of data and their suitability for testing the proposed methods: the Egg and Olympus ensembles are widely used in the literature for benchmarking purposes (with the Olympus ensemble being the more complex of the two), while the real field provides a practical application scenario for the methods developed in this work.

Simulations of the Egg and Olympus ensembles to generate the associated priors were performed using the JutulDarcy.jl simulator ([Møyner, 2025](#)).

4.2.1 Egg ensemble

Originally introduced as a deterministic model in a study aimed at exploring the reasons behind reservoir flooding problems and identifying conditions for optimal solutions ([Zandvliet *et al.*, 2007](#)), a stochastic version was later proposed as a benchmark in a robust waterflooding optimization problem ([Essen *et al.*, 2009](#)). This reservoir consists of 18,533 active cells with dimensions of 8x8x4 m. The arrangement of these cells forms an egg-shaped grid, hence the name ([Jansen *et al.*, 2014](#)).

4.2.1.1 Realizations

The standard Egg ensemble comprises a single deterministic realization and 100 permeability realizations of a channelized reservoir. In addition to eight injectors labeled INJECT1 through INJECT8, each realization includes four vertical producer wells labeled PROD1 through PROD4, which produce oil and water but not gas. [Table 4.1](#) shows some of the properties shared by the realizations.

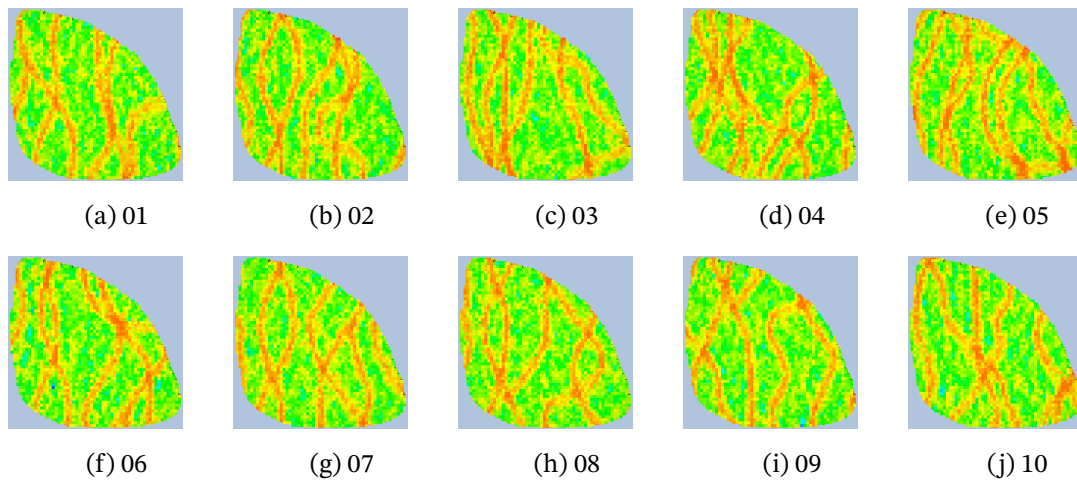
Table 4.1 – Some geometry and fluid properties of the Egg ensemble.

Property	Value	Unit
Rock Compressibility	0.0	1/Pa
Water Compressibility	1.0×10^{-10}	1/Pa
Oil Dynamic Viscosity	5.0×10^{-3}	Pa·s
Water Dynamic Viscosity	1.0×10^{-3}	Pa·s
Well Diameter	20	cm
Simulation Time	3600	days

Adapted from [Jansen *et al.* \(2014\)](#).

In [figure 4.1](#), renderings of permeability along the x-axis in some of Egg’s realizations are depicted, with colors assigned according to a logarithmic scale. To enhance visualization, the z-axis was magnified fivefold.

Figure 4.1 – Top view of selected realizations 1-10 from the Egg ensemble. Renderings were produced using the open-source software ResInsight.



Source: [Filho *et al.* \(2025\)](#).

4.2.2 Olympus ensemble

Launched in early 2017, the Olympus reservoir model was designed as a benchmark environment to problems concerning control optimization and reservoir management within the framework of the Integrated Systems Approach for Petroleum Production 2 research consortium ([Fonseca *et al.*, 2018](#); [Fonseca *et al.*, 2020](#)). In what follows, the depiction of Olympus mirrors the one outlined by [Fonseca *et al.* \(2020\)](#), with adjustments made where deemed necessary.

Inspired by an oil field in the North Sea, this synthetic reservoir is approximately rectangular, measuring 9x3 km, with a depth of 50 m divided into 16 layers. The 3D grid comprises cells with dimensions of approximately 50x50x3 m, totaling 341,728 cells, of which

192,750 are active. The perimeter of the reservoir is defined by a fault, with six additional smaller faults present within it. Furthermore, Olympus features two distinct zones separated by an impermeable shale layer: The upper zone contains fluvial channel sands embedded in floodplain shales, while the lower zone consists of alternating layers of coarse, medium, and fine sands that are inclined relative to the general structural dip of the field. The reservoir includes four facies types in different layers (see [table 4.2](#)), and geological properties, such as porosity, permeability, and the net-to-gross ratio, were generated using standard geostatistical techniques for the different facies types. In the absence of sufficient early-stage data, no porosity/permeability ratio was used. Permeabilities in the x and y directions were identical, while in the z direction, it corresponded to 10% of that in the x direction.

Table 4.2 – Types and distribution range intervals of facies properties in Olympus.

Type	Presence zones	NG	Porosity	Permeability (μm^2)
Channel sand	Upper	0.8-1	0.2-0.35	0.4-1
Shale	Upper, perimeter	0.0	0.03	0.001
Coarse sand	Lower	0.7-0.9	0.2-0.3	0.15-0.4
Medium sand	Lower	0.75-0.95	0.1-0.2	0.075-0.15
Fine sand	Lower	0.9-1	0.05-0.1	0.01-0.05

Source: [Fonseca et al. \(2020\)](#).

Based on available exploration well records, it was determined that the oil/water contact depth was 2092 m, with an in-situ hydrostatic pressure of 205 bar. The initial water saturation distribution was modeled using capillary pressure curves. Different capillary pressure curves were assigned to different facies types. Therefore, changes in the facies resulted in different initial water saturation distributions. The reservoir's fluid was similar to oil without dissolved gas, and some of its properties are summarized in [table 4.3](#).

Table 4.3 – Olympus' fluid properties.

Property	Value	Unit
Rock compressibility	1.42×10^{-10}	1/Pa
Water compressibility	3.97×10^{-10}	1/Pa
Oil dynamic viscosity	$2.8-3.5 \times 10^{-3}$	Pa·s
Water dynamic viscosity	0.398×10^{-3}	Pa·s
Well diameter	19	cm
Simulation Time	7380	days

Source: Adapted from [Fonseca et al. \(2020\)](#).

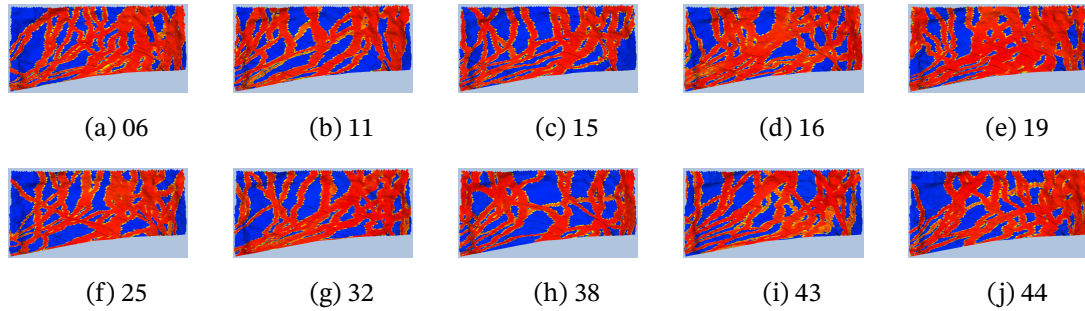
4.2.2.1 Realizations

A set of 50 realizations was generated by varying facies through random seed changes, while keeping consistent mesh geometry, faults, and oil-water contact. Uncertain properties

such as facies, porosity, permeability, NG ratio, initial water saturation distribution, and transmissibility within faults were considered.

By first generating a high-fidelity base model with approximately 5 million grid cells, 5 wells were drilled into this base case model and synthetic records were generated for each of these wells. These records were then used to constrain the generation of 50 high-fidelity realizations to capture uncertainty. Each was upscaled for flow simulations using the flow-based upscaling method, featuring a total of 18 wells in each realization, of which 11 are producers and 7 are injectors. Each realization includes 7 injector wells, labeled INJ-1 to INJ-7, and 11 producer wells, labeled PROD-1 to PROD-11. Producers produce oil and water, but not gas. Figure 4.2 illustrates the permeability in the x direction for selected realizations, depicted with colors on a logarithmic scale and a fivefold exaggeration of the z -axis.

Figure 4.2 – Top view of selected realizations from the OLYMPUS ensemble. Renderings were produced using the open-source software ResInsight.



Source: Filho *et al.* (2025).

4.2.3 Real field

Given that most parameters characterizing the properties of the carbonate reservoir in question cannot be disclosed due to ongoing confidentiality agreements with industrial partners, only essential information is provided.

The real field is a mature reservoir located in the Brazilian Post-Salt, with a production history spanning over 20 years. It is comprised of 8 injector wells, labeled I1 to I8, and 16 producer wells, labeled P1 to P16. Producers produce oil and water, but not gas.

An ensemble of 200 realizations was generated from observed production data, with a total of 510 simulation time steps of roughly 30 days each. Importantly, observed data has a time vector that is not identical to the simulation's but is a subset of it.

4.2.4 Field Oil Production Totals

The field oil production total (FOPT) is a measurement commonly used to compare ensembles since groundtruth and realizations are each represented by a single value: how

much oil was produced within a given period. ?? shows the cumulative distribution function (CDF) of field oil production total (FOPT) of the case studies ensembles. FOPT values are normalized by the maximum across all a case study's realizations.

Figure 4.3a shows that the Egg ensemble has a relatively narrow, roughly uniform range of FOPT values, suggesting that the priors should be reasonable representatives of the groundtruths, especially those that are closer to the median of the distribution.

On the other hand, figure 4.3b shows that the Olympus ensemble has a much wider (about 10 times that of the Egg ensemble), albeit still roughly uniform, range of FOPT values. The picked groundtruths (realizations 5, 10, etc.) are spread within the 5-95th percentiles, with half of them quite distant from the median. Since this reservoir's FOPTs thin out at the percentile ranges of 0-5% and 90-100%, it would be expected that groundtruths picked from these ranges would be more difficult to match, as they are less likely to be represented in the prior ensemble. This effect is further aggravated by the large range of FOPT values in the Olympus ensemble, which is roughly 10 times larger than that of the Egg ensemble.

Finally, ?? shows the real field's ensemble's CDF of FOPT, with the observed datum circled. Since the time vector of realizations go well beyond the observed period, they would become more likely to have a FOPT greater than the reference. To avoid this, FOPT values presented refer only to the observed period, and are normalized by the maximum across all realizations. This ensemble has a relatively wide range of FOPT values, with the observed datum being very close to the 50th percentile. However, the distribution quite lopsided, with a very populated 5-80th percentile range and a thin 0-5% and 80-100% range. As seen in the Olympus case, a broad range of FOPT values coupled with poorly-represented tails can lead to inconsistent posterior coverages and MSEs.

4.3 Evaluation workflow and parameters

4.3.1 Evaluation workflow

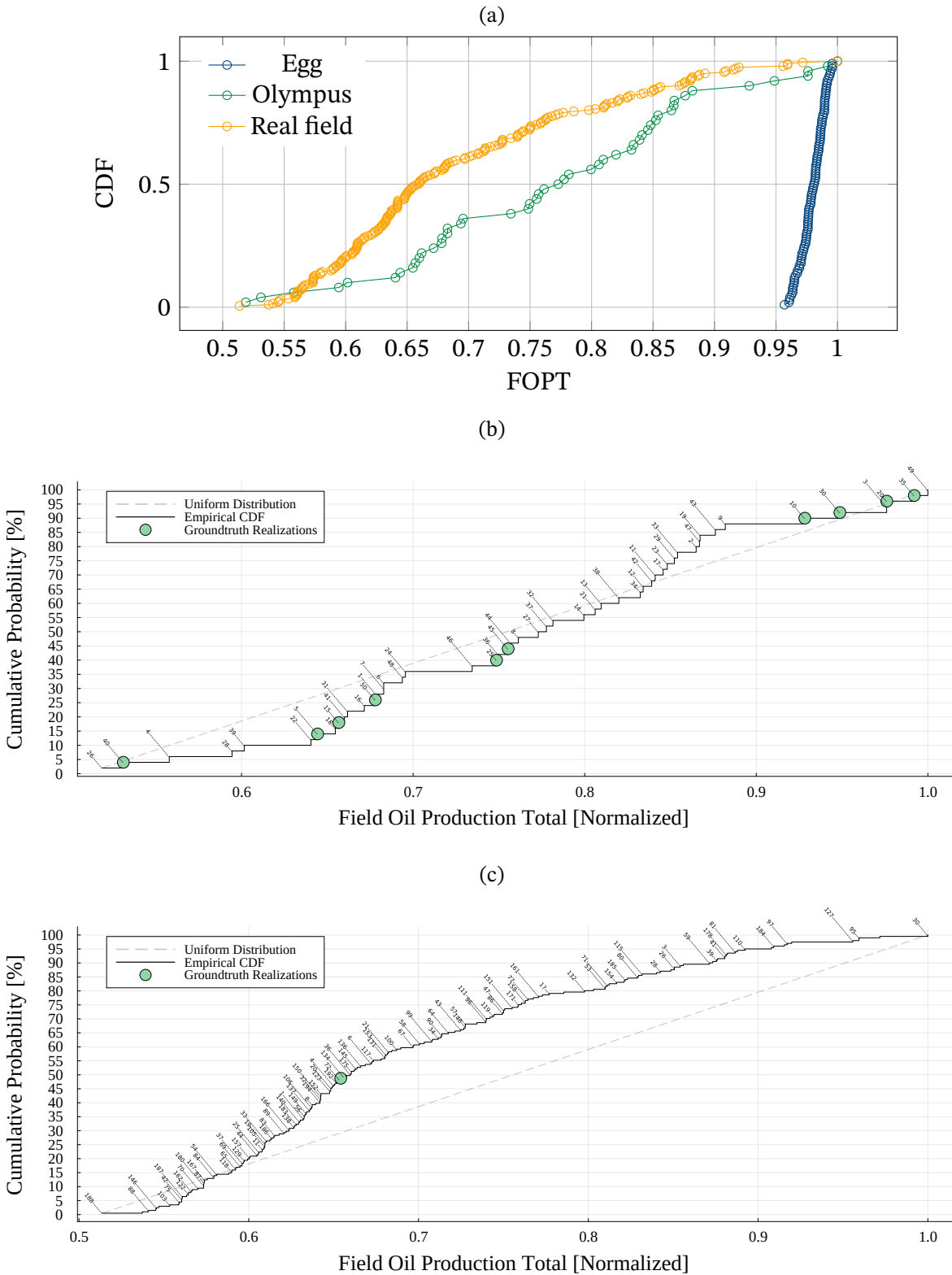
While the outline of both DSI-ESMDA and the proposed Gated DSI-ESMDA methods have been presented in chapter 3, it remains to be explained how the error covariance matrix $\mathbf{C}_e$ is constructed, what parameters are used, and what metrics will be used to evaluate the performance of the methods. Figure 4.4 illustrates the workflow of the evaluation process.

First, all data is normalized to the range $[0, 1]$ using the minimum and maximum values of the groundtruth production data, as follows:

$$\tilde{\mathbf{d}}_{j,f} = \frac{\mathbf{d}_{j,f} - \min(\mathbf{d}_f)}{\max(\mathbf{d}_f) - \min(\mathbf{d}_f)} \quad (4.1)$$

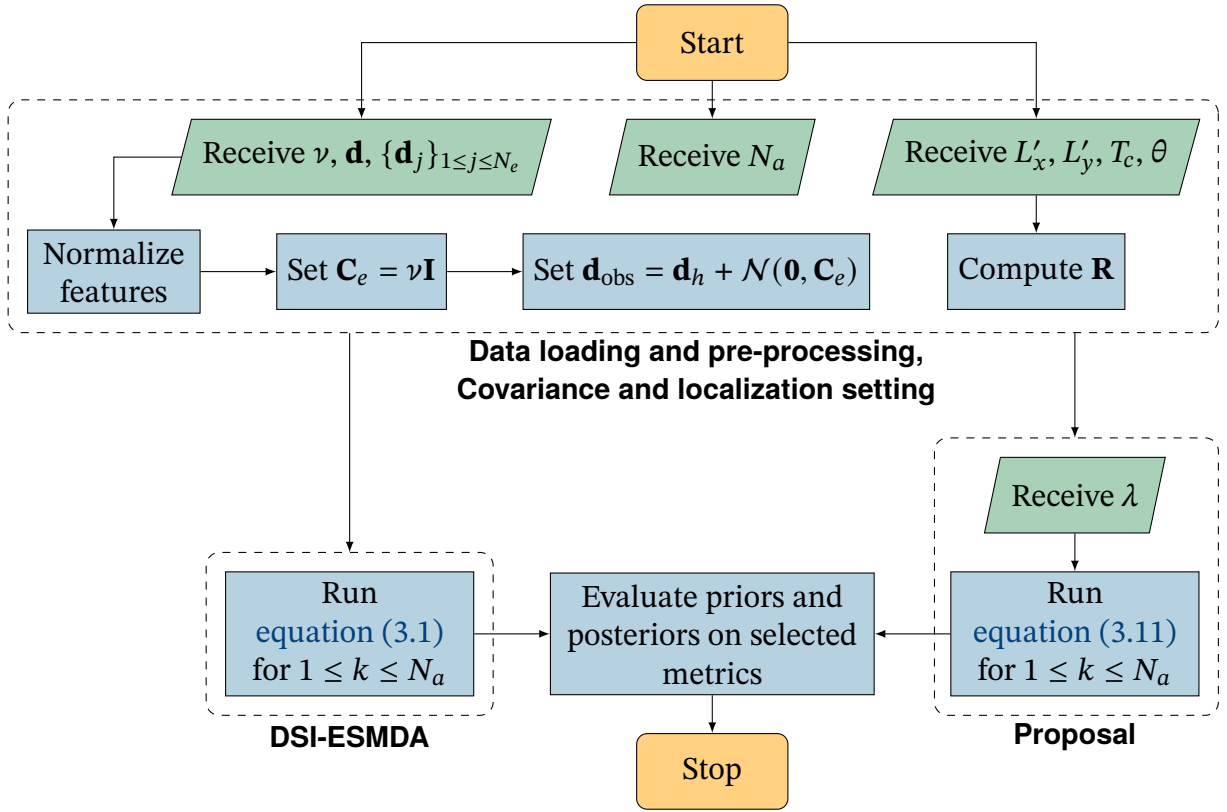
$$\tilde{\mathbf{d}}_f = \frac{\mathbf{d}_f - \min(\mathbf{d}_f)}{\max(\mathbf{d}_f) - \min(\mathbf{d}_f)}, \quad (4.2)$$

Figure 4.3 – Distribution of FOPT values for each realization in the (a) Egg, (b) Olympus, and (c) real field ensembles. Each ensemble is normalized by its maximum FOPT.



Source: Self-elaboration.

Figure 4.4 – Outline of the evaluation process.



Source: Self-elaboration.

where index j denotes the realization, f denotes the production data feature, and the tilde indicates normalization. This process is repeated for all non-constant features and realizations to improve the conditioning of the data and facilitate noise modeling, as shown next.

4.3.2 Noise modeling

The error covariance matrix $\mathbf{C}_e$ is assumed to be diagonal, with its non-zero entries uniformly sampled from $(0, 1]$ and scaled by a constant $\nu > 0$. This constant is named the noise magnitude and is a hyperparameter of the Evaluation. However, ν is only means to easier explore the effects of different noise magnitudes on the performance of the methods, and not an essential parameter of the methods themselves.

Next, observed data is constructed by adding noise to the normalized groundtruth data restricted to the history period:

$$\tilde{\mathbf{d}}_{\text{obs}} = \tilde{\mathbf{d}}_h + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_e). \quad (4.3)$$

Since real field data already contains noise, this step is skipped for that case, and observed data is simply the normalized groundtruth data restricted to the history period.

Finally, the methods are applied to the pre-processed data, and their priors and posteriors are evaluated using the metrics described in [subsection 4.3.3](#).

4.3.3 Evaluation metrics

Both the prior and posterior ensembles are evaluated on the history and forecast periods using the following metrics ([Emerick, 2025](#), sections 9.4.2 and 9.4.4):

- observation coverage (OC),
- mean squared error (MSE),
- data-mismatch objective function (denoted O).

OC measures the percentage of observed data that falls within the range of an ensemble, while MSE quantifies the average squared difference between the observed and predicted data. DMOF is a metric that quantifies the discrepancy between observed and predicted data. Mathematically, each metric is defined as follows:

$$OC_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1}(\min(\mathbf{Y})_i \leq d_{h,i} \leq \max(\mathbf{Y})_i), \quad (4.4)$$

$$OC_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1}(\min(\mathbf{Y})_i \leq d_{v,i} \leq \max(\mathbf{Y})_i), \quad (4.5)$$

$$MSE(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \bar{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \bar{\mathbf{y}}), \quad \mathbf{C} = \Delta \mathbf{Y} \Delta \mathbf{Y}^\top + \mathbf{C}_e \quad (4.6)$$

$$O(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}), \quad (4.7)$$

where

- $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{N_e}]$ denotes an ensemble (posterior or prior),
- N_h is the number of data points in the history period,
- N_v is the number of data points in the validation period,
- $\bar{\mathbf{y}}$ is the mean of the ensemble,
- $\max(\mathbf{Y})_i$ is the maximum value in the ensemble at datum i ,
- $\min(\mathbf{Y})_i$ is the minimum value in the ensemble at datum i ,
- $\Delta \mathbf{Y}$ is the deviation of the predicted data from its mean (see [equation \(3.3\)](#)),
- $\mathbb{1} : \{0, 1\} \rightarrow \{0, 1\}$ returns 1 if the condition in its argument is true and 0 otherwise.

Both history and validation splits of groundtruth $\mathbf{d}$ are used to compute observation coverages, while MSE and misfit exist only for the history period. Despite the similarity between the MSE and the misfit, the latter is computed for each realization in the ensemble, while the former is computed for the entire ensemble. In this work, the misfit will be presented as an average over all realizations in the ensemble, with an associated standard deviation.

4.3.4 Evaluation parameters

While a straightforward application of DSI-ESMDA requires knowledge of the error covariance matrix (addressed in [subsection 4.3.2](#)) and the number of Kalman-like iterations N_a , the amount of parameters to tune balloon if localization is utilized. The proposed Gated DSI-ESMDA method introduces an additional hyperparameter, the gating intensity λ . In practice, the gating is introduced after the standard DSI-ESMDA has been applied and tuned, making the tuning of λ incremental and independent of the other parameters.

[Table 4.4](#) compares the time-related simulation parameters and features across reservoirs, while [table 4.5](#) compares the localization-related parameters. Finally, [table 4.6](#) lists the main hyperparameters: number of iterations, noise magnitude, and gating intensity.

Experiments were performed on a small array of noise magnitudes and gating intensities, with the number of iterations fixed at $N_a = 1$ for all experiments.

Table 4.4 – Time-related simulation parameters per reservoir.

Reservoir	Number of realizations	Number of time steps	Time step size	Simulation life		History split
Egg	100	120	30 days	3600 days	(9.8 years)	60%
Olympus	49	20	1 calendar year	7305 days	(20 years)	60%
Real field	200	510	1 calendar month*	12785 days	(35 years)	60%**

Source: Self-elaboration.

*: The very first time step, with 274 days, is the only exception.

**: Real data time vectors aren't synchronized across features, so 60% of a feature may lead to a shorter time scope than another feature.

Table 4.5 – Localization-related simulation parameters per reservoir.

Reservoir	Critical length in x	Critical length in y	Critical time	Anisotropy angle
Egg	100	100	3000 days	0°
Olympus	100	100	7200 days	0°
Real field	-	-	10500 days	0°

Source: Self-elaboration.

Note: Critical lengths are in amount of gridblocks, and are therefore dimensionless.

4.4 Chapter summary

Key takeaways

Table 4.6 – Evaluation hyperparameters.

Parameter	Symbol	Value(s)	Unit
Kalman-like iterations	N_a	1	-
Noise magnitude	ν	{0.001, 0.01}	-
Gating intensity	λ	{1, 2, 4, 8, 16}	-

Source: Self-elaboration.

5 Results

5.1 Chapter outline

Aims and structure

5.1.1 Egg ensemble

Results for the standard DSI-ESMDA are detailed in [table 5.1](#), while results for the proposed Gated DSI-ESMDA under a mild gating intensity are shown in [table 5.2](#).

[Table 5.1](#) shows that DSI-ESMDA posteriors had a higher posterior coverage during the history period when localization was applied, in exchange for a slightly higher data mismatch average and standard deviation. Indeed, averaging across groundtruths, localization increased the posterior coverage from 35.4% ($\pm 0.84\%$) to 52.7% ($\pm 2.11\%$), while the average data mismatch increased from 0.496 (± 0.005) to 0.555 (± 0.02). It is also clear that localization tends to increase posterior coverage in the validation period, but it may also decrease it, as seen in the groundtruth 30 case. While localization did not decrease posterior MSE, values being close to 1 indicate that posterior can be considered good ($MSE \leq 1$) or at least fair ($1 < MSE \leq 2$) ([Emerick, 2025](#), section 9.4.2.2).

It is known ([??](#), appendix) that the expected mismatch is 0.5 for randomized maximum likelihood samples. Since the localization did not move the average mismatch significantly away from 0.5, it is reasonable to say that the increase in posterior coverage is overall beneficial.

From [table 5.2](#), it can be seen that the proposed Gated DSI-ESMDA method, across groundtruths, has an average posterior coverage in the history period of 41.00% ($\pm 1.76\%$) and an average mismatch of 0.513 (± 0.008) when localization isn't applied, versus 57.3% ($\pm 1.25\%$) and 0.598 (± 0.01) when it is. The increase in history coverages relative to those obtained by the DSI-ESMDA under the same conditions is significant, although so is the increase in the data mismatch. In the validation period, it is clear that localization has a positive effect on the posterior coverage, and that the proposed method posterior coverages tend to be slightly higher than those obtained by the DSI-ESMDA. Localization did not tend to decrease the posterior MSE, which remained good or fair. This was expected, since the same behavior was observed in the standard DSI-ESMDA.

The following short-hands are used in the tables and figures of this section: Loc.: Localization is applied; Pr.: Prior ensemble; Po.: Posterior ensemble; Avg.: Average value; Std. Dev.: Standard deviation.

Table 5.1 – DSI-ESMDA results for the Egg ensemble.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation		Pr.	Po.	Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.			Pr.	Po.	Pr.	Po.
10	No	0.92	0.34	1.00	1.00	1.03	0.96	9.63	0.50	5.05	5.1×10^{-3}
	Yes	0.92	0.48	1.00	1.00	1.03	0.96	9.63	0.55	5.05	0.03
20	No	0.93	0.36	1.00	0.98	0.97	0.93	6.93	0.49	3.32	4.8×10^{-3}
	Yes	0.93	0.52	1.00	1.00	0.97	0.93	6.93	0.53	3.32	0.02
30	No	0.87	0.36	0.93	1.00	0.99	0.94	16.77	0.49	7.01	4.6×10^{-3}
	Yes	0.87	0.53	0.93	0.90	0.99	0.94	16.77	0.57	7.01	0.03
40	No	0.94	0.35	1.00	1.00	1.01	0.96	8.62	0.50	4.79	5.0×10^{-3}
	Yes	0.94	0.52	1.00	1.00	1.01	0.96	8.62	0.55	4.79	0.02
50	No	0.92	0.35	1.00	0.90	1.00	0.94	10.70	0.50	5.29	5.2×10^{-3}
	Yes	0.92	0.53	1.00	1.00	1.00	0.94	10.70	0.55	5.29	0.03
60	No	0.92	0.36	0.99	0.94	0.99	0.94	9.13	0.49	5.14	4.7×10^{-3}
	Yes	0.92	0.53	0.99	0.98	0.99	0.94	9.13	0.55	5.14	0.03
70	No	0.91	0.34	1.00	1.00	1.03	0.95	8.62	0.50	4.08	5.6×10^{-3}
	Yes	0.91	0.53	1.00	1.00	1.03	0.95	8.62	0.55	4.08	0.02
80	No	0.91	0.36	1.00	0.90	1.01	0.94	26.01	0.50	19.20	5.3×10^{-3}
	Yes	0.91	0.56	1.00	1.00	1.01	0.94	26.01	0.58	19.20	0.04
90	No	0.87	0.36	0.98	0.96	1.01	0.95	43.29	0.50	31.62	4.9×10^{-3}
	Yes	0.87	0.55	0.98	1.00	1.01	0.95	43.29	0.59	31.62	0.05
100	No	0.93	0.35	1.00	1.00	0.98	0.93	7.25	0.49	3.45	4.7×10^{-3}
	Yes	0.93	0.52	1.00	1.00	0.98	0.93	7.25	0.53	3.45	0.02

Source: Self-elaboration.

Figure 5.1 summarizes results for the Egg ensemble by averaging posterior coverage and MSE across groundtruths for different gating intensities, with null gating corresponding to standard DSI-ESMDA.

Gating seems to produce effects similar to those of localization, especially when its intensity is increased. In the history period, there is a clear trend of increasing posterior coverage with increasing gating intensity, at the cost of increased MSE. Importantly, there was little effect on validation coverage, but it did tend to increase relative to the prior regardless of gating intensity. This is likely an effect of the prior ensemble being a good representative of the groundtruths, as the FOPT CDF suggested.

5.1.2 Olympus ensemble

Results for the standard DSI-ESMDA are presented in table 5.3, while results for the proposed Gated DSI-ESMDA are shown in table 5.4.

Compared to the Egg case, priors of the Olympus ensemble presented extremely high data mismatches (to the point of producing expuriose deviations), which is a testament to the large spread in range this ensemble has.

Table 5.2 – Gated DSI-ESMDA ($\lambda = 2$) results for the Egg ensemble.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation		Pr.	Po.	Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.			Pr.	Po.	Pr.	Po.
10	No	0.92	0.41	1.00	1.00	1.03	0.97	9.63	0.52	5.05	0.01
	Yes	0.92	0.57	1.00	1.00	1.03	0.97	9.63	0.60	5.05	0.02
20	No	0.93	0.43	1.00	1.00	0.97	0.94	6.93	0.51	3.32	0.01
	Yes	0.93	0.58	1.00	1.00	0.97	0.94	6.93	0.58	3.32	0.02
30	No	0.87	0.39	0.93	0.99	0.99	0.95	16.77	0.51	7.01	0.01
	Yes	0.87	0.55	0.93	0.90	0.99	0.95	16.77	0.61	7.01	0.03
40	No	0.94	0.42	1.00	1.00	1.01	0.97	8.62	0.53	4.79	0.01
	Yes	0.94	0.57	1.00	1.00	1.01	0.97	8.62	0.60	4.79	0.02
50	No	0.92	0.43	1.00	0.90	1.00	0.95	10.70	0.51	5.29	0.01
	Yes	0.92	0.59	1.00	1.00	1.00	0.95	10.70	0.60	5.29	0.02
60	No	0.92	0.41	0.99	0.95	0.99	0.94	9.13	0.51	5.14	0.01
	Yes	0.92	0.57	0.99	0.96	0.99	0.95	9.13	0.60	5.14	0.02
70	No	0.91	0.40	1.00	1.00	1.03	0.96	8.62	0.52	4.08	0.01
	Yes	0.91	0.57	1.00	1.00	1.03	0.97	8.62	0.60	4.08	0.02
80	No	0.91	0.40	1.00	0.93	1.01	0.95	26.01	0.50	19.20	0.01
	Yes	0.91	0.59	1.00	1.00	1.01	0.95	26.01	0.60	19.20	0.03
90	No	0.87	0.38	0.98	0.94	1.01	0.96	43.29	0.51	31.62	9.3×10^{-3}
	Yes	0.87	0.56	0.98	0.99	1.01	0.97	43.29	0.61	31.62	0.04
100	No	0.93	0.43	1.00	1.00	0.98	0.94	7.25	0.51	3.45	9.3×10^{-3}
	Yes	0.93	0.58	1.00	1.00	0.98	0.94	7.25	0.58	3.45	0.02

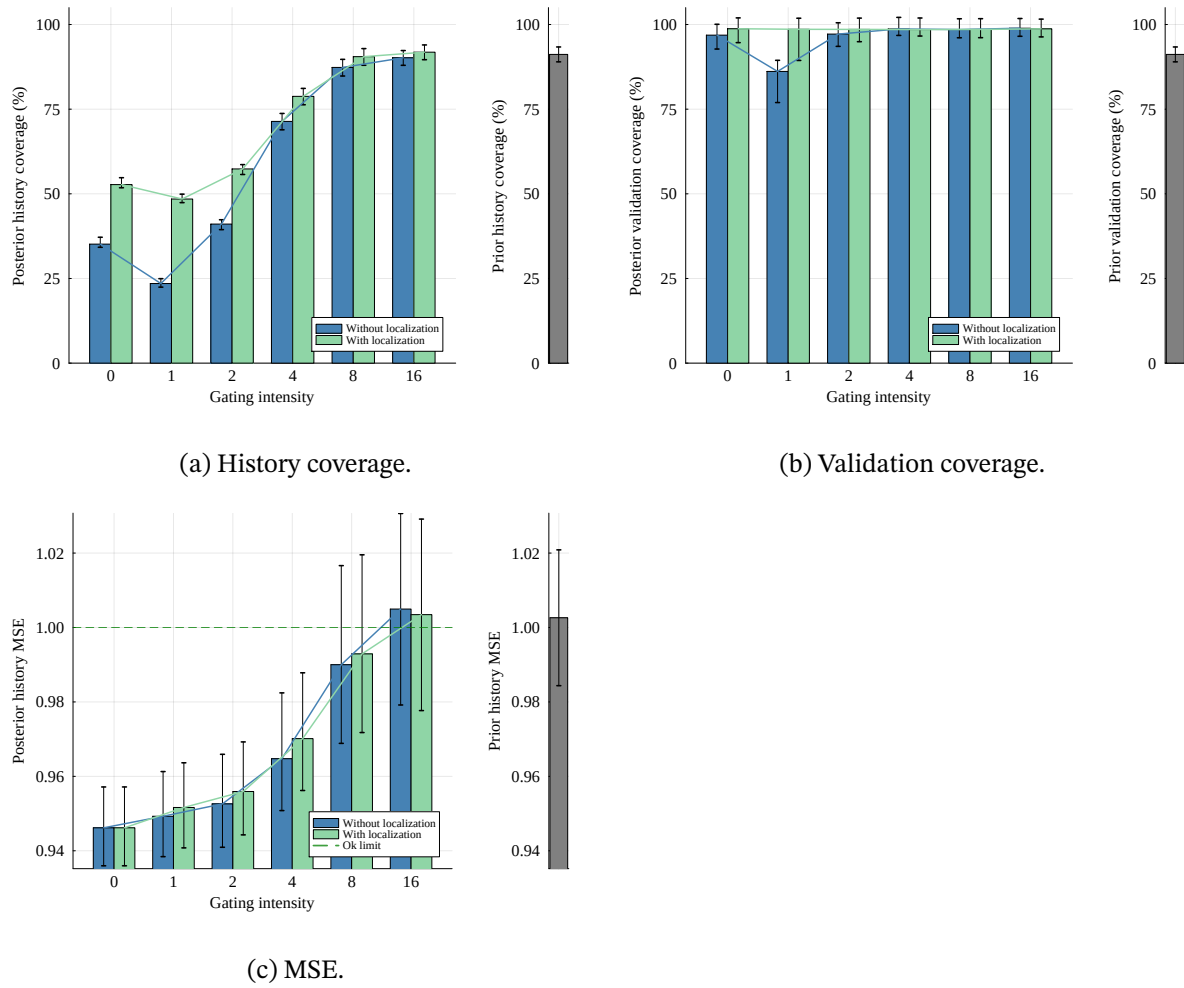
Source: Self-elaboration.

Table 5.3 confirms that localization had a pronounced positive effect on posterior coverage, both in the history and validation periods, albeit at the risk of incurring high data mismatches. Averaging across groundtruths, localization increased the posterior coverage from 40.00% ($\pm 5.01\%$) to 64.80% ($\pm 5.75\%$) in the history period, and from 47.5% ($\pm 18.53\%$) to 87.8% ($\pm 4.92\%$) in the validation period, while the average data mismatch increased from 0.897 (± 0.257) to 3.340 (± 5.267). Localization was also effective in decreasing the posterior MSE, even managing to fair biased MSEs (groundtruths 15, 45).

Table 5.4 shows that the trend is similar for the proposed approach. Averaging across groundtruths, localization increased the posterior coverage from 28.90% ($\pm 6.54\%$) to 61.7% ($\pm 6.72\%$) in the history period, and from 37.40% ($\pm 18.17\%$) to 88.50% ($\pm 4.90\%$) in the validation period, while the average data mismatch increased from 0.869 (± 0.246) to 3.337 (± 5.263). As with the standard DSI-ESMDA, localization was also effective in decreasing the posterior MSE, managing to fair the same biased MSEs (groundtruths 15, 45).

Figure 5.2 summarizes results for the Olympus ensemble by averaging posterior coverage and MSE across groundtruths for different gating intensities, with null gating corresponding to standard DSI-ESMDA.

Figure 5.1 – Average posterior coverage and MSE values found in the Egg ensemble.



Source: Self-elaboration.

As with the Egg case, intensifying the gating tends to increase the posterior coverage while trading off MSE. Unlike the Egg case, however, MSE values of gated DSI-ESMDA are on average consistently higher than those of standard DSI-ESMDA. Furthermore, validation coverage varied wildly, with localization greatly boosting the metric for null or mild gating.

The larger deviations and ranges on the average MSE and coverages are likely due, at least in some part, to the sampling of groundtruth realizations: FOPT-wise, half of them come from the tails of the distribution, which are weakly represented in the prior ensemble. In order to explore the effects of gating on a more typical groundtruth, Appendix C shows the prior and posterior distributions for realization 27, which is the ensemble's median FOPT. The posterior is reasonably well-matched to the groundtruth.

Table 5.3 – DSI-ESMDA results for the Olympus ensemble.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History Pr.	History Po.	Validation Pr.	Validation Po.	Pr.	Po.	Avg. Pr.	Avg. Po.	Std. Dev. Pr.	Std. Dev. Po.
5	No	0.89	0.28	0.96	0.26	2.90	2.70	2.2×10^7	1.42	2.7×10^7	0.01
	Yes	0.89	0.66	0.96	0.76	2.90	2.29	2.2×10^7	17.80	2.7×10^7	18.21
10	No	0.93	0.40	0.97	0.59	2.10	1.83	1.6×10^3	0.98	1.7×10^3	0.02
	Yes	0.93	0.59	0.97	0.85	2.10	1.56	1.6×10^3	1.30	1.7×10^3	0.29
15	No	0.96	0.41	0.97	0.26	2.71	2.05	3.3×10^2	1.10	2.7×10^2	0.02
	Yes	0.96	0.70	0.97	0.89	2.71	1.75	3.3×10^2	1.49	2.7×10^2	0.33
20	No	0.83	0.42	0.88	0.63	1.41	1.11	62.06	0.62	43.78	0.02
	Yes	0.83	0.57	0.88	0.86	1.41	1.09	62.06	0.72	43.78	0.06
25	No	0.96	0.36	0.97	0.27	1.81	1.64	9.5×10^9	0.89	1.2×10^{10}	0.02
	Yes	0.96	0.71	0.97	0.91	1.81	1.41	9.5×10^9	5.49	1.2×10^{10}	4.78
30	No	0.89	0.43	0.96	0.78	1.37	1.15	4.2×10^2	0.63	4.3×10^2	0.01
	Yes	0.89	0.60	0.96	0.87	1.37	1.10	4.2×10^2	0.75	4.3×10^2	0.09
35	No	0.89	0.42	0.92	0.45	1.73	1.49	2.9×10^2	0.81	2.7×10^2	0.01
	Yes	0.89	0.59	0.92	0.92	1.73	1.34	2.9×10^2	1.05	2.7×10^2	0.18
40	No	0.96	0.45	0.97	0.56	1.52	1.41	6.4×10^2	0.77	5.0×10^2	0.01
	Yes	0.96	0.72	0.97	0.88	1.52	1.24	6.4×10^2	1.76	5.0×10^2	0.74
45	No	0.92	0.39	0.93	0.34	2.37	2.07	5.7×10^2	1.10	5.2×10^2	0.01
	Yes	0.92	0.64	0.93	0.93	2.37	1.82	5.7×10^2	1.99	5.2×10^2	0.73
50	No	0.95	0.45	0.96	0.61	1.35	1.18	3.1×10^2	0.65	2.5×10^2	0.01
	Yes	0.95	0.70	0.96	0.91	1.35	1.12	3.1×10^2	1.05	2.5×10^2	0.23

Source: Self-elaboration.

5.1.3 Brazilian Post-salt Field

Results for the standard DSI-ESMDA are presented in [table 5.5](#), while results for the proposed Gated DSI-ESMDA are shown in [table 5.6](#).

Unlike the trend set by the synthetic case studies, localization did not have a significant impact on observation coverages or data mismatches in the real field case study, be it in the standard DSI-ESMDA or the proposed Gated DSI-ESMDA. However, it is clear that the gating produces better coverages as the gating intensity increases, with the best results occurring at $\lambda = 16$, the largest value tested. Importantly, this comes at the cost of a higher data mismatch and MSE, which is expected since the posterior is gaining variance.

Mild gating intensities ($\lambda = 1$, $\lambda = 2$) produce similar results to the standard DSI-ESMDA. This indicates that the gating mechanism may act as a substitute to simpler localizations on complexly correlated features, which is a significant advantage since localization tuning isn't straightforward.

5.2 Chapter summary

Table 5.4 – Gated DSI-ESMDA ($\lambda = 2$) results for the Olympus ensemble.

Groundtruth realization	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation		Pr.	Po.	Avg.	Std. Dev.		Po.
		Pr.	Po.	Pr.	Po.				Pr.	Po.	
5	No	0.89	0.16	0.96	0.14	2.90	2.71	2.2×10^7	1.37	2.7×10^7	5.7×10^{-3}
	Yes	0.89	0.66	0.96	0.76	2.90	2.26	2.2×10^7	17.80	2.7×10^7	18.21
10	No	0.93	0.29	0.97	0.47	2.10	1.84	1.6×10^3	0.96	1.7×10^3	0.01
	Yes	0.93	0.57	0.97	0.87	2.10	1.62	1.6×10^3	1.31	1.7×10^3	0.30
15	No	0.96	0.30	0.97	0.21	2.71	2.07	3.3×10^2	1.07	2.7×10^2	0.01
	Yes	0.96	0.67	0.97	0.89	2.71	1.75	3.3×10^2	1.48	2.7×10^2	0.34
20	No	0.83	0.32	0.88	0.55	1.41	1.14	62.06	0.61	43.78	0.01
	Yes	0.83	0.54	0.88	0.86	1.41	1.15	62.06	0.74	43.78	0.04
25	No	0.96	0.25	0.97	0.18	1.81	1.65	9.5×10^9	0.84	1.2×10^{10}	6.2×10^{-3}
	Yes	0.96	0.69	0.97	0.91	1.81	1.39	9.5×10^9	5.45	1.2×10^{10}	4.76
30	No	0.89	0.34	0.96	0.71	1.37	1.17	4.2×10^2	0.63	4.3×10^2	0.01
	Yes	0.89	0.55	0.96	0.91	1.37	1.14	4.2×10^2	0.77	4.3×10^2	0.08
35	No	0.89	0.30	0.92	0.41	1.73	1.52	2.9×10^2	0.79	2.7×10^2	0.01
	Yes	0.89	0.53	0.92	0.92	1.73	1.40	2.9×10^2	1.06	2.7×10^2	0.18
40	No	0.96	0.29	0.97	0.34	1.52	1.41	6.4×10^2	0.73	5.0×10^2	7.8×10^{-3}
	Yes	0.96	0.70	0.97	0.89	1.52	1.25	6.4×10^2	1.74	5.0×10^2	0.72
45	No	0.92	0.22	0.93	0.26	2.37	2.08	5.7×10^2	1.06	5.2×10^2	8.5×10^{-3}
	Yes	0.92	0.59	0.93	0.93	2.37	1.83	5.7×10^2	1.97	5.2×10^2	0.72
50	No	0.95	0.33	0.96	0.47	1.35	1.19	3.1×10^2	0.63	2.5×10^2	7.8×10^{-3}
	Yes	0.95	0.67	0.96	0.91	1.35	1.13	3.1×10^2	1.05	2.5×10^2	0.23

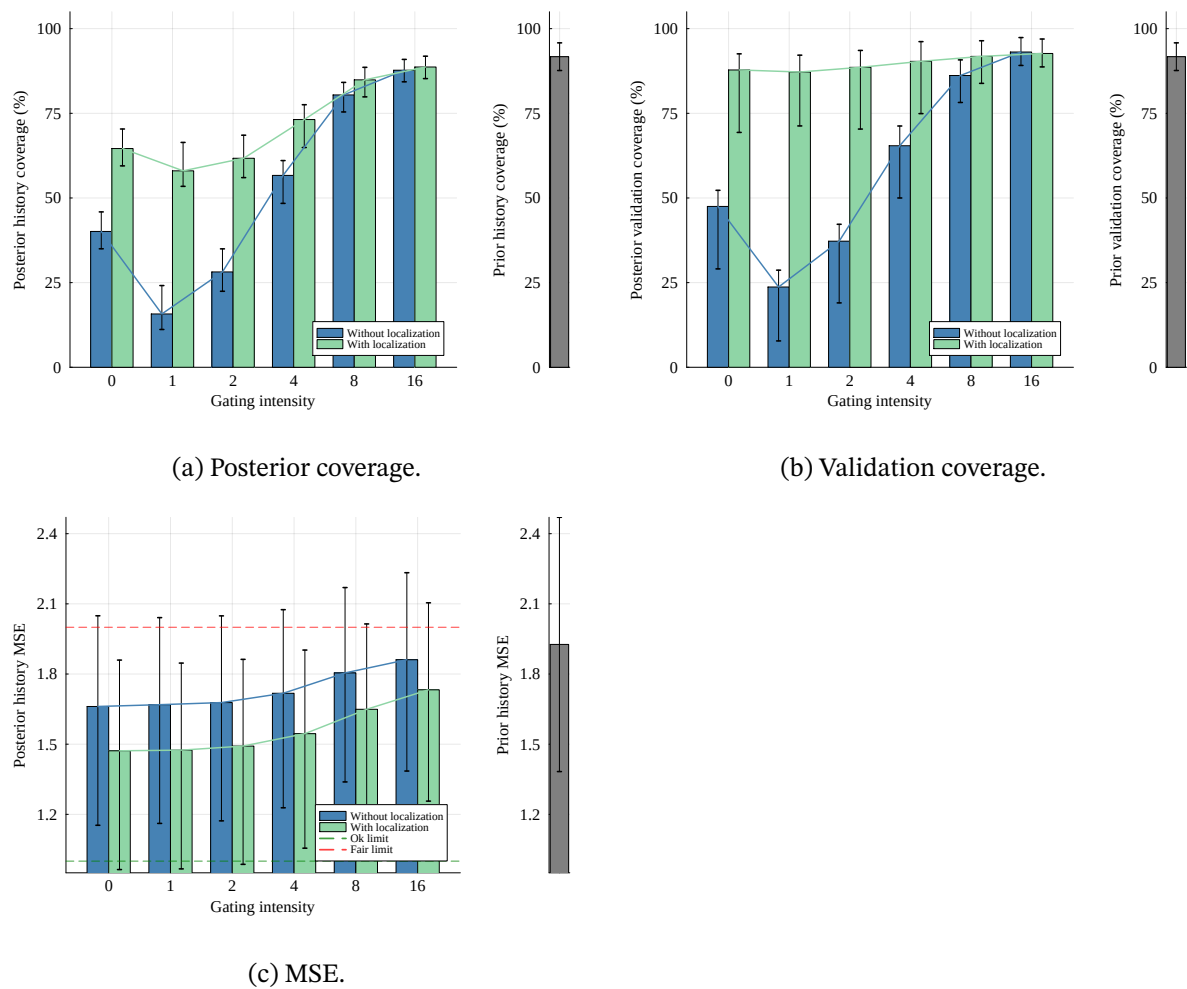
Source: Self-elaboration.

Table 5.5 – DSI-ESMDA results for the real field in the Brazilian Post-salt.

Loc.	Observation coverages				MSE		Data mismatch			
	History		Validation		Pr.	Po.	Avg.	Std. Dev.		Po.
	Pr.	Po.	Pr.	Po.				Pr.	Po.	
No	0.86	0.63	0.95	0.57	2.71	2.28	5.44	1.24	2.19	0.02
Yes	0.86	0.63	0.95	0.57	2.71	2.28	5.44	1.24	2.19	0.02

Source: Self-elaboration.

Figure 5.2 – Summary of results for the Olympus ensemble. Posterior coverage and MSE are averaged across groundtruths.



Source: Self-elaboration.

Key takeaways

Table 5.6 – Gated DSI-ESMDA results for the real field in the Brazilian Post-salt with varying gating.

Gating intensity	Loc.	Observation coverages				MSE		Data mismatch			
		History		Validation				Avg.		Std. Dev.	
		Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.	Pr.	Po.
$\lambda = 1$	No	0.86	0.53	0.95	0.52	2.71	2.32	5.44	1.21	2.19	0.01
	Yes	0.86	0.54	0.95	0.55	2.71	2.32	5.44	1.21	2.19	0.01
$\lambda = 2$	No	0.86	0.56	0.95	0.55	2.71	2.36	5.44	1.27	2.19	0.02
	Yes	0.86	0.57	0.95	0.58	2.71	2.36	5.44	1.27	2.19	0.02
$\lambda = 4$	No	0.86	0.63	0.95	0.64	2.71	2.44	5.44	1.52	2.19	0.04
	Yes	0.86	0.64	0.95	0.66	2.71	2.44	5.44	1.53	2.19	0.05
$\lambda = 8$	No	0.86	0.73	0.95	0.74	2.71	2.55	5.44	2.15	2.19	0.19
	Yes	0.86	0.73	0.95	0.75	2.71	2.55	5.44	2.16	2.19	0.19
$\lambda = 16$	No	0.86	0.80	0.95	0.81	2.71	2.66	5.44	3.10	2.19	0.57
	Yes	0.86	0.80	0.95	0.81	2.71	2.66	5.44	3.12	2.19	0.57

Source: Self-elaboration.

6 Discussion

It's tough to make predictions, especially about the future.

Yogi Berra

6.1 Chapter outline

Aims and structure

6.2 Summary of Key Findings

6.3 Interpretation of Findings

6.4 Evaluation of Existing Theories and Models

6.5 Limitations and Future Research

6.6 Chapter summary

Key takeaways

7 Conclusion

7.1 Summary of Findings

Can a gating mechanism be introduced into DSI-ESMDA to selectively modulate the influence of observations based on their match quality?

Answer.

Does the proposed innovation gating reduce variance loss during assimilation and produce more robust uncertainty envelopes in the prediction phase without degrading the data match?

Answer.

How does the performance of Gated DSI-ESMDA compare to standard DSI-ESMDA across synthetic and real reservoir case studies?

Answer.

7.2 Contribution to Knowledge

7.3 Practical Implications

7.4 Limitations

7.5 Future Research

7.6 Closing Statement

References

- DONOHU, D. De-noising by soft-thresholding. **IEEE Transactions on Information Theory**, v. 41, n. 3, p. 613–627, 1995. DOI [10.1109/18.382009](https://doi.org/10.1109/18.382009). Cit. on pp. 9 and 16.
- DONOHU, D. L.; JOHNSTONE, I. M. Ideal spatial adaptation by wavelet shrinkage. **Biometrika**, v. 81, n. 3, p. 425–455, 09 1994. ISSN 0006-3444. DOI [10.1093/biomet/81.3.425](https://doi.org/10.1093/biomet/81.3.425). Available at: <https://doi.org/10.1093/biomet/81.3.425>. Cit. on pp. 9 and 16.
- EMERICK, A. A. **Ensemble Data Assimilation Applied to Geological Reservoir Models**. 1. ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, 2025. ISBN 978-65-88763-29-2. Available at: <https://publicacoesup.petrobras.com.br/peld/catalog/book/54>. Cit. on pp. 17, 32, and 35.
- ESSEN, G. M. van; ZANDVLIET, M. J.; HOF, P. M. J. van den; BOSGRA, O. H.; JANSEN, J. D. Robust Waterflooding Optimization of Multiple Geological Scenarios. **SPE Journal**, v. 14, n. 01, p. 202–210, 03 2009. DOI [10.2118/102913-PA](https://doi.org/10.2118/102913-PA). Cit. on p. 25.
- FILHO, J. O. d. A. L.; PAULA, T. T. d.; PASTRANA, M. A.; NOZ, D. M. M.; SANTOS, M. G.; FORTALEZA, E. L. F. Robust adaptive model aggregation for well production forecasting. **SPE Journal**, v. 30, n. 07, p. 3984–4006, 07 2025. ISSN 1086-055X. DOI [10.2118/226207-PA](https://doi.org/10.2118/226207-PA). Available at: <https://doi.org/10.2118/226207-PA>. Cit. on pp. 26 and 28.
- FONSECA, R.-M.; ROSSA, E.; EMERICK, A.; HANEA, R.; JANSEN, J.-D. Overview of the olympus field development optimization challenge. *In: ECMOR XVI - 16th European Conference on the Mathematics of Oil Recovery*. [S.l.: s.n.], 2018. v. 2018, p. 1–10. DOI [10.3997/2214-4609.201802246](https://doi.org/10.3997/2214-4609.201802246). Cit. on p. 26.
- FONSECA, R. M.; ROSSA, E. D.; EMERICK, A. A.; HANEA, R. G.; JANSEN, J. D. Introduction to the special issue: Overview of olympus optimization benchmark challenge. **Computational Geosciences**, v. 24, p. 1933–1941, 2020. DOI [10.1007/s10596-020-10003-4](https://doi.org/10.1007/s10596-020-10003-4). Cit. on pp. 16, 26, and 27.
- FORTALEZA, E. L.; GOMES, L. M.; de A. Limaverde Filho, J. O.; de Campos, M. C.; LONGHI, L. G. S.; de Lima, M. L.; TOGNETTI, E. S. Model-free adaptive filter to mitigate actuator wear. **ISA Transactions**, v. 129, p. 493–504, 2022. ISSN 0019-0578. DOI <https://doi.org/10.1016/j.isatra.2022.02.026>. Available at: <https://www.sciencedirect.com/science/article/pii/S0019057822000829>. Cit. on pp. 9, 16, 23, and 50.
- JANSEN, J. D.; FONSECA, R. M.; KAHROBAEI, S.; SIRAJ, M. M.; ESSEN, G. M. van; HOF, P. M. J. van den. The egg model - a geological ensemble for reservoir simulation.

- Geoscience Data Journal**, v. 1, n. 2, p. 192–195, 2014. DOI [10.1002/gdj3.21](https://doi.org/10.1002/gdj3.21). Cit. on pp. 16, 25, and 26.
- JIANG, S.; DURLOFSKY, L. J. History matching for geological carbon storage using data-space inversion with spatio-temporal data parameterization. **International Journal of Greenhouse Gas Control**, v. 134, p. 104124, 2024. ISSN 1750-5836. DOI <https://doi.org/10.1016/j.ijggc.2024.104124>. Cit. on p. 18.
- JIANG, S.; HUI, M.-H.; DURLOFSKY, L. J. Data-space inversion with a recurrent autoencoder for naturally fractured systems. **Frontiers in Applied Mathematics and Statistics**, Volume 7 - 2021, 2021. ISSN 2297-4687. DOI [10.3389/fams.2021.686754](https://doi.org/10.3389/fams.2021.686754). Cit. on p. 18.
- LIMA, M. M.; EMERICK, A. A.; ORTIZ, C. E. P. Data-space inversion with ensemble smoother. **Computational Geosciences**, v. 24, n. 3, p. 1179–1200, Jun 2020. ISSN 1573-1499. DOI [10.1007/s10596-020-09933-w](https://doi.org/10.1007/s10596-020-09933-w). Available at: <https://doi.org/10.1007/s10596-020-09933-w>. Cit. on pp. 9, 14, 18, and 20.
- MØYNER, O. Jutuldarcy.jl - a fully differentiable high-performance reservoir simulator based on automatic differentiation. **Computational Geosciences**, Springer, v. 29, n. 30, 2025. Cit. on p. 25.
- OLIVER, D. S.; CHEN, Y. Recent progress on reservoir history matching: a review. **Computational Geosciences**, v. 15, n. 1, p. 185–221, Jan 2011. ISSN 1573-1499. DOI [10.1007/s10596-010-9194-2](https://doi.org/10.1007/s10596-010-9194-2). Available at: <https://doi.org/10.1007/s10596-010-9194-2>. Cit. on p. 15.
- SUN, W.; DURLOFSKY, L. J. A new data-space inversion procedure for efficient uncertainty quantification in subsurface flow problems. **Mathematical Geosciences**, v. 49, n. 6, p. 679–715, 2017. DOI [10.1007/s11004-016-9672-8](https://doi.org/10.1007/s11004-016-9672-8). Available at: <https://doi.org/10.1007/s11004-016-9672-8>. Cit. on pp. 9, 14, and 18.
- XIAO, C.; ZHANG, S.; MA, X.; ZHOU, T.; HOU, T.; CHEN, F. Deep-learning-generalized data-space inversion and uncertainty quantification framework for accelerating geological co2 plume migration monitoring. **Geoenergy Science and Engineering**, v. 224, p. 211627, 2023. ISSN 2949-8910. DOI <https://doi.org/10.1016/j.geoen.2023.211627>. Cit. on p. 18.
- ZANDVLIET, M. J.; BOSGRA, O. H.; JANSEN, J. D.; HOF, P. M. J. van den; KRAAIJE-VANGER, J. F. B. M. Bang-bang control and singular arcs in reservoir flooding. **Journal of Petroleum Science and Engineering**, v. 58, n. 1, p. 186–200, 2007. ISSN 0920-4105. DOI [10.1016/j.petrol.2006.12.008](https://doi.org/10.1016/j.petrol.2006.12.008). Cit. on p. 25.

Appendices

Appendix A – Academic Productions

The master’s program in Mechanical Engineering at University of Brasília (UnB) requires the completion of a minimum of 360 hours of coursework (24 credits) and at least one publication in a national or international conference/journal. This appendix exhibits the academic productions resulting from the research work developed by the author during the course of the master’s degree.

A.1 Disciplines Completed

The author has completed the following disciplines, which are summarized in [table A.1](#).

Table A.1 – Summary of the disciplines completed by the author during their master’s degree.

Start	End	Code	Name	Hours	Credits
10/2024	05/2026	PCMEC0073	SEMINÁRIO 1 EM CIÊNCIAS MECÂNICAS	60h	4
10/2024	02/2025	PCMEC0076	PROCESSAMENTO DE SINAIS E SISTEMAS MECÂNICOS LINEARES	60h	4
10/2024	02/2025	PCMEC2158	MÉTODOS NUMÉRICOS EM CIÊNCIAS MECÂNICAS	60h	4
10/2024	02/2025	PCMEC0091	TÉCNICAS NÃO LINEARES EM CONTROLE*	60h	4
03/2025	08/2025	PCMEC0147	VIBRAÇÕES MECÂNICAS	60h	4
03/2025	07/2025	PCMEC0192	TERMODINÂMICA CLÁSSICA	60h	4
08/2025	12/2025	PCMEC0180	ESTÁGIO DE DOCÊNCIA EM CIÊNCIAS MECÂNICAS 1	60h	4
08/2025	12/2025	PPGEE2249	APRENDIZADO DE MÁQUINA	60h	4
03/2026	07/2025	PPGEE2166	PROCESSOS ESTOCÁSTICOS	60h	4
Total (minimum: 360h / 24 credits)				540h	36

Source: Self-elaboration.

*: This discipline was completed during the author’s undergraduate studies, but is a graduate-level discipline.

A.2 Publications

Papers the author has published during their master’s degree are summarized in [table A.2](#).

Table A.2 – Summary of the author’s publications during their master’s degree.

Type	Title	Event	Submission
National conference paper	Custos de Operação e Manutenção Associados ao Serviço de Flexibilidade das Hidrelétricas	XI Simpósio Brasileiro de Sistemas Elétricos (SBSE)	March 17th, 2025
National conference paper	Error-Based Active Disturbance Rejection Control for Parkinson’s Disease Tremor Suppression in Wrist Considering Upper Limb Dynamics	XXVI Congresso Brasileiro de Automática (CBA)	May 2nd, 2026
International conference paper	Nonlinear Innovation Gating for Variance Preservation in DSI-ESMDA Forecasting	20th European Conference on the Mathematics of Geological Reservoirs (ECMOR)	July 11th, 2026
Total (minimum: 1)			3

Source: Self-elaboration.

A.3 Software

During their master’s degree, the author participated in a research project with Axia Energia (then Eletrobras Furnas), which produced a software for the estimation of the opportunity cost of hidreletric flexibility. Details of the software are provided in [table A.3](#).

Table A.3 – Summary of the software developed by the author during their master’s degree.

Title	Cálculo do Custo de Oportunidade de Usinas Hidrelétricas
Summary	Library for estimating the opportunity cost associated with the flexibility of hydroelectric plants, through deterministic models, to be used as a decision support tool for electric grid operators.
Company concerns being addressed	Renewable energies, predictibility, cost assessment
Creation date	20/10/2025
Hash algorithm	SHA-512
Hash	3a24c572e979a5f8897f4787a33fac173d7fc3d5a197bb79d640fc26423744d640f1eec72e082973b85af7c5f555bc20294b47a5e05479d2f06cf9ad5e5c3da5
Language	Python
Application fields	Energy politics, Technology and energy (energy generation)
Kind of program	Library, support tool, simulation and modeling, techno-scientific applications
Authors	Thiago Tomás de Paula , Yan Tavares de Oliveira, Matheus Bawden Silverio de Castro, Matheus Teixeira de Sousa, Eduardo Bezerra Rufino Ferreira Paiva, José Oniram de Aquino Limaverde Filho, William Humberto Cuéllar Sánchez, Thiago Felício de Souza, Eugênio Libório Feitosa Fortaleza

Source: Self-elaboration.

Appendix B – Deriving the Innovation Gating Parameter

This appendix provides a self-contained derivation of the alpha parameter used in the innovation gating mechanism, as introduced in [equation \(3.12\)](#) of [chapter 3](#). The derivation follows closely the work of [Fortaleza *et al.* \(2022\)](#), from where the gating mechanism was inspired.

Let $Y \in \mathbb{R}$ be a random variable representing the output of a given process in time, and Y_I the observation of Y . It is assumed that the observation is affected by a Gaussian noise $N \sim \mathcal{N}(0, \sigma_n^2)$, such that

$$Y_I = Y + N. \quad (\text{B.1})$$

Let y_I be a realization of Y_I . Denoting by y_M the prediction of y from a given model, y_O is defined to be the adjusted prediction after the observation y_I is taken into account. For the sake of simplicity, y_O is given by the following exponential moving average (EMA) relationship:

$$\begin{cases} y_O[0] = y_I[0], & k = 0, \\ y_O[k] = \alpha[k]y_I[k] + (1 - \alpha[k])y_M[k - 1], & k > 0 \end{cases} \quad (\text{B.2})$$

where $\alpha[k] \in [0, 1]$ is the smoothing factor of the EMA at time step index k .

Parameter $\alpha[k]$ represents the statistical confidence that the filter output reflects the same direction of the process output at sample k . More precisely, $\alpha[k]$ is given by the following expression:

$$\begin{aligned} \alpha[k] = & P(\text{sign}(Y_M[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) \\ & - P(\text{sign}(Y_M[k - 1] - Y_I[k]) \neq \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})), \end{aligned} \quad (\text{B.3})$$

where $Y_M[k - 1]$ is an observation of Y_M with mean $\mu_{Y_M[k-1]}$ and Y_I an observation of Y_I with mean $\mu_{Y_I[k]}$. Here $\text{sign}(x)$ represents the sign function, which is defined as:

$$\text{sign}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases} \quad (\text{B.4})$$

[Equation \(B.3\)](#) can be rewritten as follows:

$$\alpha[k] = 2P(\text{sign}(Y_M[k - 1] - Y_I[k]) = \text{sign}(\mu_{Y_M[k-1]} - \mu_{Y_I[k]})) - 1. \quad (\text{B.5})$$

Defining a random variable Y_{MI} whose observation at time k have mean and variance given by

$$\mu_{Y_{MI}[k]} = |\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|, \quad (\text{B.6})$$

$$\sigma_{Y_{MI}[k]}^2 = \sigma_{Y_M[k-1]}^2 + \sigma_{Y_I[k]}^2 - 2\text{Cov}(Y_M[k-1], Y_I[k]), \quad (\text{B.7})$$

where $\text{Cov}(Y_M[k-1], Y_I[k])$ is the covariance between $Y_M[k-1]$ and $Y_I[k]$, [equation \(B.5\)](#) can be expressed as follows:

$$\alpha[k] = 2P(Y_{MI}[k] > 0) - 1. \quad (\text{B.8})$$

In order to find a practical implementation of $\alpha[k]$, [equation \(B.8\)](#) is analyzed under the following technical assumptions.

Assumption B.1. The random variable $Y_{MI}[k]$ can be approximated by a normal distribution with mean $\mu_{Y_{MI}[k]}$ and variance $\sigma_{Y_{MI}[k]}^2$.

Assumption B.2. Both high frequency and random behavior of the measurement noise implies that the observations $Y_M[k-1]$ and $Y_I[k]$ can be considered approximately statistically independent of each other. Mathematically, $\text{Cov}(Y_M[k-1], Y_I[k]) \approx 0$.

Assumption B.3. The variance of the model's prediction is much smaller than the variance of the measured observation. Mathematically, $\sigma_{Y_M[k-1]}^2 \ll \sigma_{Y_I[k]}^2$.

Assumption B.4. By considering an additive zero-mean measurement noise, the samples $y_M[k-1]$ and $y_I[k]$ can be used to estimate $\mu_{Y_M[k-1]}$ and $\mu_{Y_I[k]}$, respectively.

From [assumption B.1](#), [equation \(B.8\)](#) can be rewritten as:

$$\begin{aligned} \alpha[k] &= 2P(\mathcal{N}(\mu_{Y_{MI}[k]}, \sigma_{Y_{MI}[k]}^2) > 0) - 1 \\ &= 2P(\mathcal{N}(0, \sigma_{Y_{MI}[k]}^2) \leq \mu_{Y_{MI}[k]}) - 1, \end{aligned} \quad (\text{B.9})$$

which can be expressed in terms of the Gauss error function $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-x^2} dx$:

$$\begin{aligned} \alpha[k] &= \text{erf}\left(\frac{\mu_{Y_{MI}[k]}}{\sqrt{2\sigma_{Y_{MI}[k]}^2}}\right) \\ &= \text{erf}\left(\frac{|\mu_{Y_M[k-1]} - \mu_{Y_I[k]}|}{\sqrt{2[\sigma_{Y_M[k-1]}^2 + \sigma_{Y_I[k]}^2 - 2\text{Cov}(Y_M[k-1], Y_I[k])]}}\right). \end{aligned} \quad (\text{B.10})$$

Applying [assumptions B.2 to B.4](#), [equation \(B.10\)](#) becomes

$$\alpha[k] = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma_{Y_I}^2[k]}} \right) = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\sqrt{2\sigma_n^2[k]}} \right). \quad (\text{B.11})$$

Since knowledge of the measurement noise variance $\sigma_n^2[k]$ is not always available in practice, a fixed $\lambda \geq 1$ is introduced to scale the noise standard deviation. Thus, the final expression for $\alpha[k]$ is given by

$$\alpha[k] = \operatorname{erf} \left(\frac{|y_M[k-1] - y_I[k]|}{\lambda \sqrt{2\sigma_n^2[k]}} \right) \quad (\text{B.12})$$

which is equivalent to [equation \(3.12\)](#) in [chapter 3](#).

